

The Unit Circle

Constructing the Unit Circle

Using the provided graph, follow the procedure below to construct your own unit circle.

PROCEDURE: Constructing the Unit Circle

Complete these steps to construct a unit circle for quick reference.

Step 1: Draw the xy -coordinate axes, a large circle centered at the origin with radius of one unit, and mark the coordinate $(-1,0)$ as π as a reminder that $180^\circ = \pi$ radians.

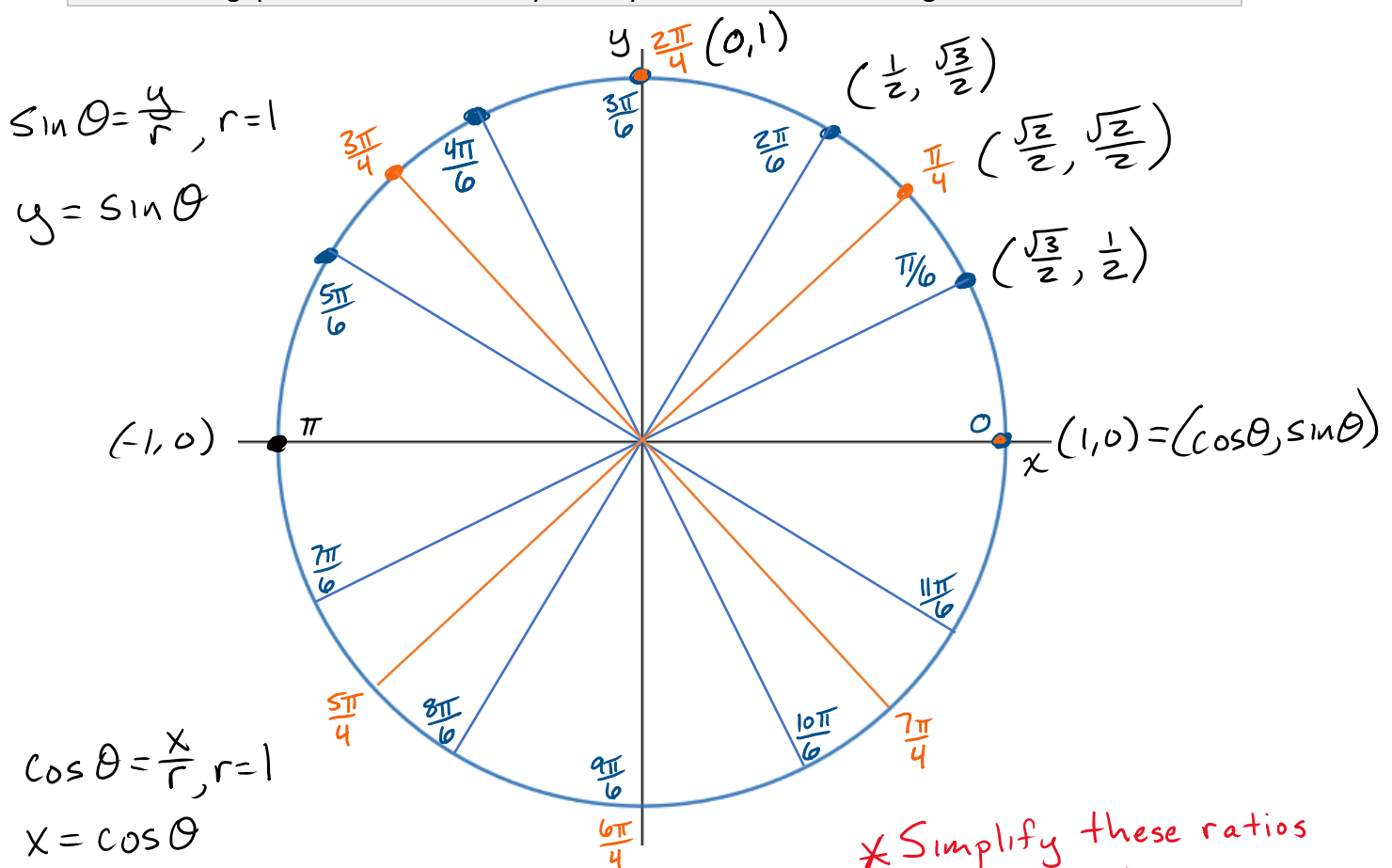
Step 2: Since $\pi/6$ is an important angle, mark coordinates to divide the upper semi-circle into 6 equal pieces. Carefully construct chords passing through your points and the origin.

Step 3: Count off the marked coordinates along the edge of the circle, $\frac{\pi}{6}, \frac{2\pi}{6}, \frac{3\pi}{6}, \frac{4\pi}{6}, \dots$

Step 4: Repeat steps 2 and 3, this time using $\pi/4$ and coordinates dividing the upper semi-circle into 4 equal segments.

Step 5: Each coordinate along the unit circle corresponds to a cosine and sine value. You should reason for yourself why this is. Label each coordinate in quadrant I using the Quadrant I Chart and the rule, $(x, y) = (\cos(\theta), \sin(\theta))$.

Step 6: (optional) Add degree measures and label the remaining coordinates along the circle using quadrant I values and symmetry. This additional labeling can wait until needed.



The Unit Circle is one of the most valuable tools provided in this text. Much like the Quadrant I Chart, the Unit Circle is reference material that will be helpful as you work through the remainder of this course. It's important to note that this circle does **NOT** need to be memorized. Instead, with a little practice, you can easily construct this valuable reference without the need for rote memorization.

Applying the Unit Circle

Up to this point, we've relied on reference angles to evaluate trigonometric functions for special angles. However, we can now use the unit circle to evaluate these expressions as an efficient and reliable alternative.

EXAMPLE 1

Evaluate each expression using the unit circle

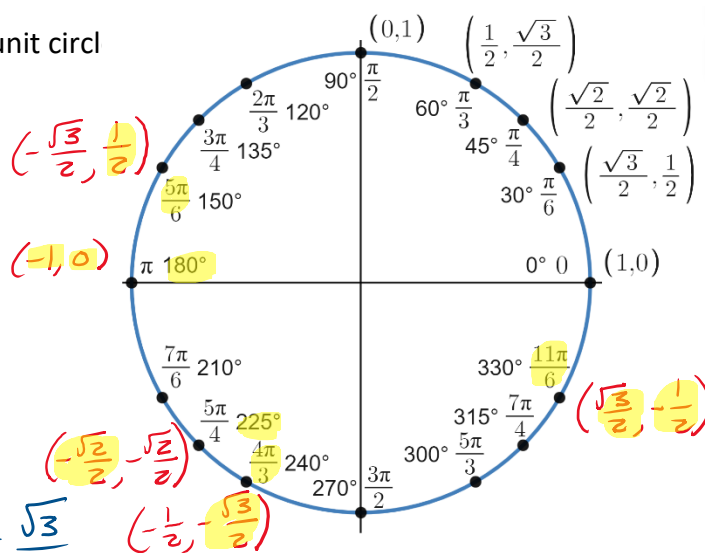
(a) $\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$
 $\uparrow$
 y-value

(b) $\cos(225^\circ) = -\frac{\sqrt{2}}{2}$
 $\uparrow$
 x-value

(c) $\tan\left(\frac{11\pi}{6}\right) = \frac{-1/2}{\sqrt{3}/2} = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$
 $\uparrow$
 y-value
 x-value

(d) $\csc\left(\frac{4\pi}{3}\right) = \frac{1}{\sin(4\pi/3)} = \frac{1}{-\sqrt{3}/2} = -\frac{2}{\sqrt{3}} = -\frac{2\sqrt{3}}{3}$
 $\uparrow$
 y-value

(e) $\cot(180^\circ) = \frac{-1}{0}$, undefined
 $\uparrow$
 x-value
 y-value



Sometimes we'll be working with special angles that yield exact values but are not typically included on the unit circle. We may be able to evaluate these by simply adding or subtracting 360° or 2π as many times as necessary to obtain a coterminal angle that does appear along the unit circle.

EXAMPLE 2

Evaluate each expression using the unit circle

$$(a) \sin\left(-\frac{5\pi}{6}\right) = \sin\left(\frac{7\pi}{6}\right) = -\frac{1}{2}$$

$$-\frac{5\pi}{6} + \frac{12\pi}{6} = \frac{7\pi}{6}$$

$(-1, 0)$

$$(b) \cos(-690^\circ) = \cos(30^\circ) = \frac{\sqrt{3}}{2}$$

$$-690 + 360 = -330$$

$$-330 + 360 = 30$$

$(\frac{\sqrt{3}}{2}, \frac{1}{2})$

$$(c) \tan(5\pi) = \tan(\pi) = \frac{0}{-1} = 0$$

$$5\pi - 2\pi = 3\pi$$

$$3\pi - 2\pi = \pi$$

$$(d) \csc\left(\frac{17\pi}{6}\right) = \csc\left(\frac{5\pi}{6}\right) = \frac{1}{\sin(5\pi/6)} = \frac{1}{1/2} = 2$$

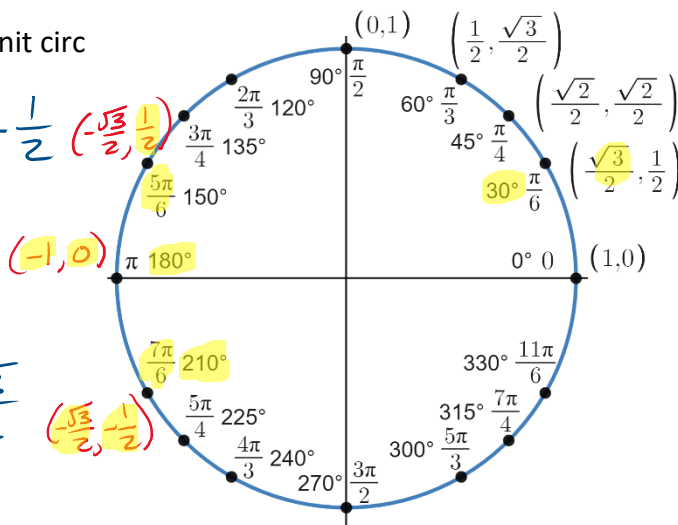
$$\frac{17\pi}{6} - \frac{12\pi}{6} = \frac{5\pi}{6}$$

$$(e) \sec(1290^\circ) = \sec(210^\circ) = \frac{1}{\cos(210^\circ)} = \frac{1}{-\sqrt{3}/2} = -\frac{2}{\sqrt{3}} = -\frac{2\sqrt{3}}{3}$$

$$1290 - 360 = 930$$

$$930 - 360 = 570$$

$$570 - 360 = 210$$



The unit circle can also help us evaluate trigonometric functions given information about other trigonometric function values.

EXAMPLE 3

Assume $0 \leq \theta < 2\pi$. Using the provided unit circle, answer each of the following questions.

- (a) Given $\cos(\theta) = -\frac{1}{2}$, find the values for the other five trigonometric functions.

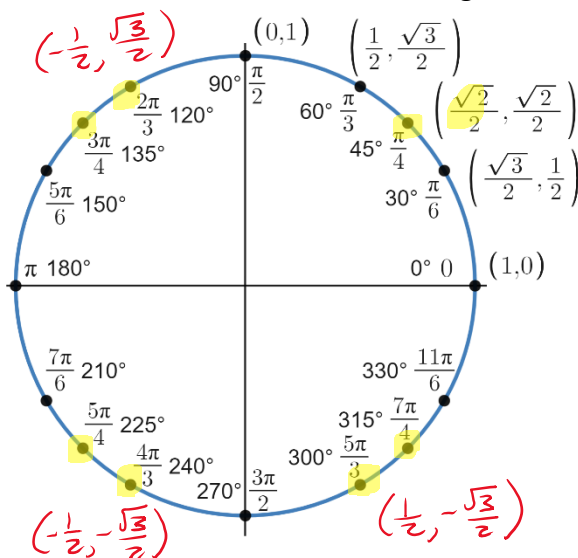
$$\sec \theta = -2$$

$$\sin \theta = \pm \frac{\sqrt{3}}{2} \quad (y)$$

$$\csc \theta = \pm \frac{2}{\sqrt{3}} = \pm \frac{2\sqrt{3}}{3}$$

$$\tan \theta = \pm \sqrt{3} \quad (y/x)$$

$$\cot \theta = \pm \frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$$



- (b) Given $\sin(\theta) = -\frac{\sqrt{3}}{2}$ and $\cos(\theta) > 0$, find $\tan(\theta)$, if possible.

$$\hookrightarrow \theta = \frac{5\pi}{3}, \tan\left(\frac{5\pi}{3}\right) = \frac{-\sqrt{3}/2}{1/2} = -\sqrt{3}$$

- (c) Find the exact value for θ given that $\cos^2(\theta) = \frac{1}{2}$.

$$\cos^2 \theta = \frac{1}{2}$$

$$\cos \theta = \pm \sqrt{\frac{1}{2}} = \pm \frac{\sqrt{2}}{2}$$

$$\theta = \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \right\}$$

